EX. NO: 14 DATE: 30/07/2026

WRITE A CODE USING RAW SOCKETS TO IMPLEMENT PACKET SNIFFING

AIM:

Write a code using RAW sockets to implement packet sniffing.

ALGORITHM:

1. Create a raw socket.
2. Bind the socket to the network interface.
3. Receive packets continuously.
4. Read the packet header.
5. Display the source and destination IP address.
6. Display the protocol used.
7. Continue until the user stops the program.

INPUT:

Network packets received through the network interface.

PROGRAM CODE:

```
import socket
# Create a raw socket (requires administrator/root privileges)
s = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
# Bind the socket to the network interface
s.bind(("0.0.0.0", 0))
print("packet sniffer started...")
# Continuously receive and inspect incoming packets
while True:
    packet, address = s.recvfrom(65535)
    print("Source IP: ", address[0])
    print("Packet Length: ", len(packet))
    print("-" * 30)
```

OUTPUT:

```
Packet sniffer started...
Source IP: 192.168.1.10
Packet Length: 74
------------------------
Source IP: 192.168.1.1
Packet Length: 65
```

RESULT:

Thus, a raw socket program was successfully implemented to capture & display network packets.